4. Focht, Spicer, and Fairchok (2002) report on a study of the use of duct tape to remove warts; the current standard treatment is to remove the warts through freezing. Twenty-five patients had their warts treated using freezing, and 26 patients had their warts treated using duct tape.

(a) Suppose that the researchers expected 60% of the patients treated using freezing to have their treatment successful (both in their null and alternative hypotheses). They wish to test the null hypothesis that the duct tape treatment is equally effective as the freezing treatment, vs. the alternative hypothesis that 85% of the patients treated using tape to have their treatment successful. Calculate the power for detecting an effect of this size. Use a one-sided test of size 2.5%.

(b) Under the assumptions of part (a), and assuming that an equal number of individuals will be given each treatment, determine a sample size in each group to give 80% power.

(c) Many of these children have multiple warts. Suppose that the difficulty in removing a wart on a child is influenced by properties inherent to the child, and hence all of the warts on a child might be more or less difficult to treat. Suggest an alternative study design that you expect to be more powerful, and describe how one might analyze the resulting data.